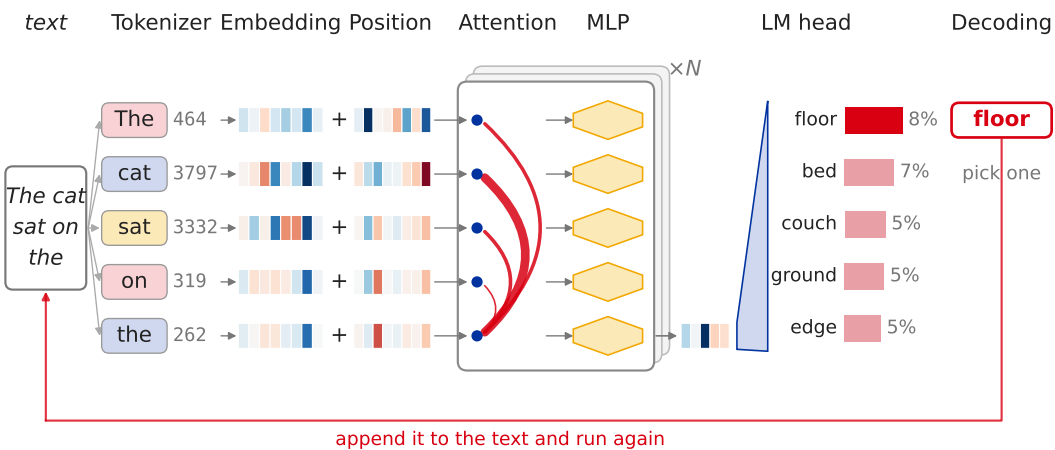
# Evaluation and Hallucination

## How do you know an answer is right?

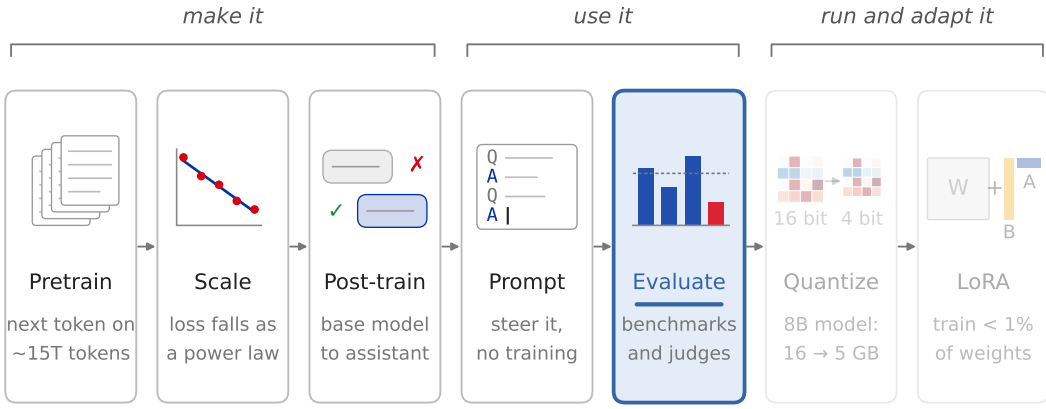
LLM chapter, session 11. Toy numbers unless a slide says otherwise.

# Where we are: one forward pass




The pass writes a fluent answer whether or not it is right. Today: telling which.

# Where we are: the life of a model




Today: the last box of "use it". Measuring a model, and catching it when it is wrong.

# Ask twice

LLM-4's two prompts, back again. This time GPT-2 *samples* at temperature 1 (LLM-6), 20 times each:

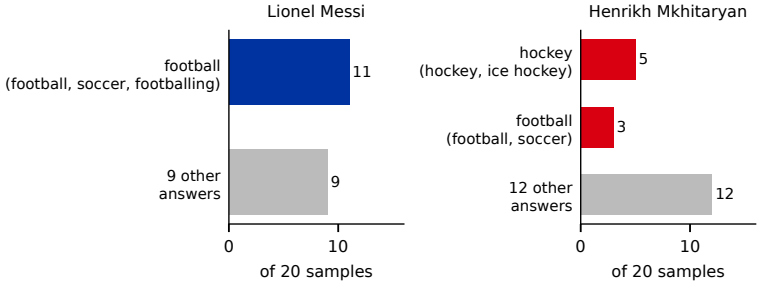
"Lionel Messi plays the sport of . . ."

"Henrikh Mkhitaryan plays the sport of . . ."

(LLM-4: GPT-2 knows the first fact, football at 35%; it misses the second, hockey first, football at 5%.)

**Predict first:** can you tell which answer to trust *without* knowing the truth, just from the 20 samples?

# Agreement is a signal




**Yes.** Grouped by meaning: Messi's 20 samples use 12 different first words, but 11 mean football. Mkhitaryan's use 16; the biggest group, hockey (both "ice" samples say "ice hockey"), holds 5, and it is wrong. (20 samples are noisy: football 2 in 20 vs 5% in LLM-4.)

A stored fact comes back the same way; a guess scatters. Sample several times and measure the agreement, counting answers that *mean* the same as one (two named methods at the end).

# Outline

Compare with a reference

Ask another model

Use a test set

Confidently wrong

# Compare with a reference

Count the words you share with a known good answer.

# Word overlap, by hand

Reference: `the cat sat on the mat`. What share of a candidate's words (1-grams), and of its word pairs (2-grams), appear in the reference? That share is **n-gram precision**.

| | **candidate** | **words** | **word pairs** |
|---|---|---|---|
| a right paraphrase | `a feline rested upon the rug` | 1/6 | 0/5 |
| close | `the cat is on the mat` | 5/6 | 3/5 |
| nonsense | `the mat sat on the cat` | **6/6** | **4/5** |

Each candidate word counts at most as often as it appears in the reference ("the" twice).
**BLEU** (translation) is built on this precision: how much of the candidate is in the reference.
**ROUGE** (summaries) asks the reverse, recall: how much of the reference the candidate covers.

The nonsense sentence gets a perfect word score; the correct paraphrase gets almost nothing. Overlap measures wording, not meaning.

# What overlap is for

- It works when the answer's *wording* is fairly fixed: translation and summarisation against several reference texts, averaged over thousands of sentences. It is cheap, fast and repeatable.
- It fails for open answers: a good answer to "explain attention" can share almost no words with any reference.
- Comparing embeddings (LLM-2) instead of words catches paraphrases ("feline" near "cat"), but still cannot tell a fluent wrong answer from a right one.

For open-ended answers, someone has to read them and judge. At scale, that someone is usually another model.

# Ask another model

A model reads the answer and grades it.

# LLM-as-judge

- **Pointwise**: "Score this answer from 1 to 10." **Pairwise**: "Which of A and B is better?"
- Why use a model at all: no formula decides whether an answer is helpful or correct. A person can, but not 10,000 times a night.
- Zheng et al. (2023): strong judges (GPT-4) agreed with human preferences over 80% of the time - the same level at which two humans agree with each other.

80% agreement is good, not perfect, and it was measured on one kind of question. Check it on yours before you trust it.

# Its biases

Zheng et al. (2023) name three, plus limited reasoning:

- **Position**: preferring whichever answer is shown first (or second). Fix: ask twice with A and B swapped; count a win only if both orders agree, otherwise a tie.
- **Verbosity**: preferring the longer answer, even when it adds nothing.
- **Self-enhancement**: preferring answers written by the judge's own model family.

The same kind of habit as LLM-9's reward hacking: a model optimised against a judge will learn the judge's tastes, not just what people wanted.

# How to use a judge without lying to yourself

- **Decompose the task.** "Is this statement supported, yes or no" is a question a model answers well. "Score this answer out of 10" is not.
- **Calibrate once against humans.** Label 50 examples yourself. If the judge agrees with you much less often than people agree with each other (about 80%), its scores are decoration.
- **Use it for comparison, never as an absolute.** "0.81" means nothing. "0.81 versus 0.74 for last week's version, same judge, same questions" means something.
- **Freeze the judge.** Change the judge model or its prompt and every earlier number stops being comparable. Version it like code.

Never let the judge be the same model, with the same prompt, as the generator. You are asking it to mark its own homework.

# Use a test set

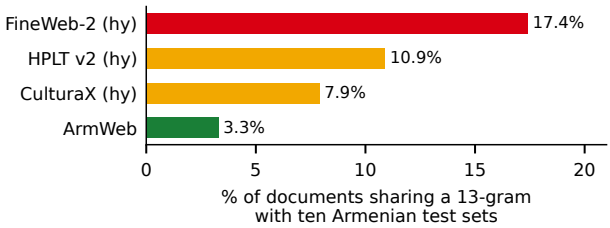
Fixed questions with known answers.

# Benchmarks

- A benchmark is a fixed set of questions with checkable answers: knowledge (MMLU, LLM-8), maths, code with unit tests, multi-step agent tasks.
- It is a *sample* of skills. A score says how the model did on those questions, in that format, scored that way - nothing more.
- Leaderboards turn a score into a target. Goodhart's law (LLM-9): once a measure becomes a target, it stops being a good measure - models get tuned to the test.

A benchmark answers "better than last month on these questions?" well, and "good at the job?" poorly.

# Contamination




Four Armenian training corpora, scanned against ten Armenian test sets for 13 words in a row copied exactly (a 13-gram). Almost all of the overlap is with the web-derived test sets; the hand-built multiple-choice sets: 0.06% or less in every corpus.

**Contamination**: test questions that are also in the training data. LLM-7's lesson: a model holds what it saw, so a contaminated item measures recall, not skill. The same study found FineWeb-2's own Armenian *test* split overlapping its *training* split.

Web-scraped test sets end up in web-scraped training sets. Prefer hand-built or newly written test questions, and check the overlap before you trust a score.

Avetisyan et al. (2026), arXiv 2609.03350 (the arm-gemma-e4b paper).

# Confidently wrong

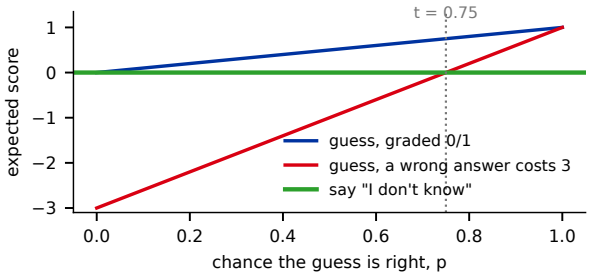
Fluent, plausible, and false.

# Hallucination

- **Factuality** errors: wrong about the world ("Mkhitaryan plays hockey" - he is a footballer).
- **Faithfulness** errors: wrong about the source you gave it - a summary that adds a claim the document never made (the RAG chapter measures this).
- Why it happens: the model always produces *some* next token (LLM-5), and a fluent wrong one looks exactly like a fluent right one. Nothing in next-token training forces "I don't know".

Kalai et al. (2025) add a second reason: our tests reward guessing. Their fix is a confidence target $t$: answer only if you are more than $t$ sure.

# Guess or abstain, by hand




A question the model gets right with probability $p$ if it guesses. "I don't know" scores 0.

**Graded 0/1**: guessing scores $p$ on average, more than 0 for any $p$. Always guess.

**A wrong answer costs 3**: guessing scores $p - 3(1-p)$. At $p = 0.5$: $-1.0$; at $p = 0.8$: $+0.2$. Guess only above 0.75.

Kalai et al.: "Answer only if you are $> t$ confident, since mistakes are penalized $t/(1-t)$ points." $t = 0.75$ gives $0.75/0.25 = 3$. Most benchmarks grade 0/1 ($t = 0$), which teaches models to guess.

Kalai et al. (2025). The paper's own example says "penalty 2" for $t = 0.75$; its formula gives 3. We use the formula.

# Catching it

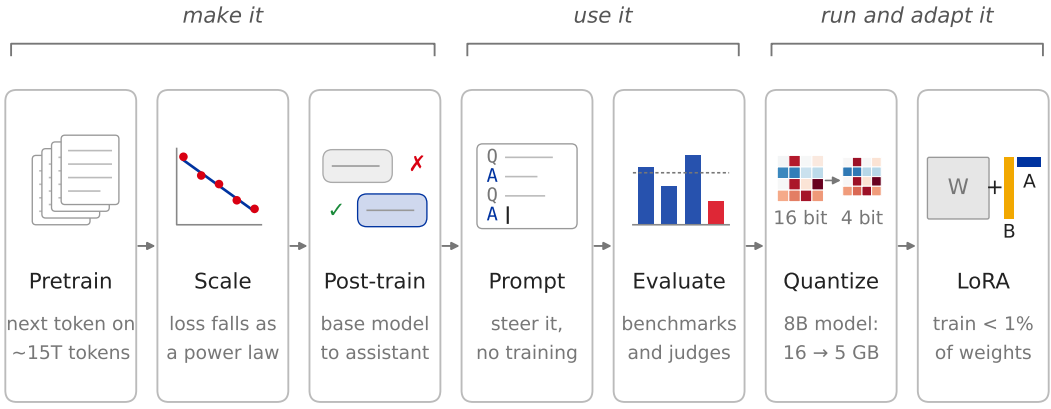
- **Ask several times** (the cold open): SelfCheckGPT (Manakul et al., 2023) samples the model again and checks each claim of the answer against every sample (with a similarity score or another model); claims the samples do not back up are flagged.
- **Count meanings, not words**: semantic entropy (Farquhar et al., 2024) groups the samples by meaning first, then measures the spread. Messi: one meaning holds 11 of 20, low spread. Mkhitaryan: the biggest holds 5, high spread - distrust it.
- **Ground it**: give the model the documents and check the answer against them. That is the next chapter.

Agreement catches guesses, not shared mistakes: a model that learned a wrong fact well repeats it every time.

# Recap

- **Reference**: word overlap (BLEU, ROUGE) is cheap and measures wording, not meaning - the nonsense sentence scored 6/6.
- **Judge**: over 80% agreement with people, with position, verbosity and self-preference biases; decompose, calibrate, compare, freeze.
- **Test set**: a sample of skills; contamination turns it into a memory test (FineWeb-2's Armenian documents: 17.4%).
- **Hallucination**: no built-in "I don't know", and 0/1 grading rewards guessing. Sampling agreement is a cheap detector.

# Next: the RAG chapter




Next (L41): give the model the documents - RAG. Then agents (L44), then LLM-12.

# Sources & further reading

- Papineni et al. (2002), "BLEU"; Lin (2004), "ROUGE".
- Zheng et al. (2023), "Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena".
- Avetisyan et al. (2026), "From Zero to Hero: An Open LLM Ecosystem for Armenian", arXiv 2609.03350 (corpus contamination).
- Kalai, Nachum, Vempala & Zhang (2025), "Why Language Models Hallucinate".
- Manakul et al. (2023), "SelfCheckGPT"; Farquhar et al. (2024), "Detecting hallucinations in large language models using semantic entropy", *Nature*.

Figures: `py_src/eval_figs.py`. Real: GPT-2's 20 samples per prompt (cached model, seed 509) and the published contamination rates. Toy: the overlap table and the guessing plot.